1.8 Systems of Linear Inequalities

Name _____ Period ____ Date _____

Algebra II 1.8 · Student sheet

I can…

- tell whether an ordered pair is a solution of a system of linear inequalities.
- graph a system of linear inequalities.
- write a system of linear inequalities from a graph or a real-life situation.

Vocabulary · fill in the missing word

System of linear inequalities — two or more linear inequalities in the same _____

Solution of a system — an ordered pair that makes every _____ true

Graph of a system — the region where the graphs of all the _____ overlap

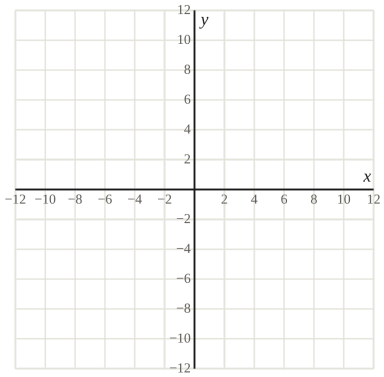
Boundary line — _____ for < or >, solid for ≤ or ≥

Part 1 · Party budget

You have $60 for pizzas ($10 each, x) and packs of drinks ($5 each, y). You need at least 2 pizzas and at least 3 packs of drinks.

(x, y)	Cost	Works? Why or why not?
(2, 3)		
(3, 6)		
(4, 5)		
(1, 8)		

System: _____



Part 2 · Notes and examples (fill in each note, then work the example)

Solution of a system: A solution of a system makes _____ inequality true.

Example 1

Tell whether each point is a solution of the system $y \geq x + 1$ and $y < -2x + 5$.

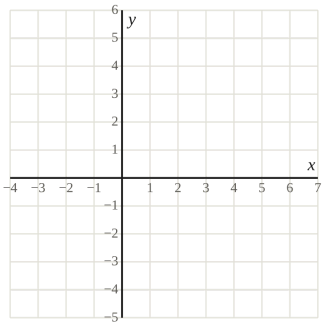
a. (1, 2) b. (3, 0)

Boundary lines: Dashed line for _____; solid line for _____.

Graph of a system: The graph of the system is the region where the shadings _____.

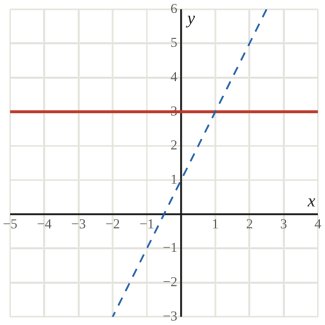
Example 2

Graph the system $y > x - 2$ and $y \le -x + 4$. Name one solution.



Example 3

The solution region lies on or below the solid line and above the dashed line. Write the system.



Example 4

You can work at most 20 hours a week babysitting ($10 per hour) and tutoring ($15 per hour). You want to earn at least $240.

Write a system and give one possible schedule.

Part 3 · Partner practice

Partner 1

Is $(-1, 4)$ a solution of $y > -x + 2$ and $y \le 3x + 8$?

Partner 2

Graph $y < 2$ and $x \ge -1$. Name one solution.